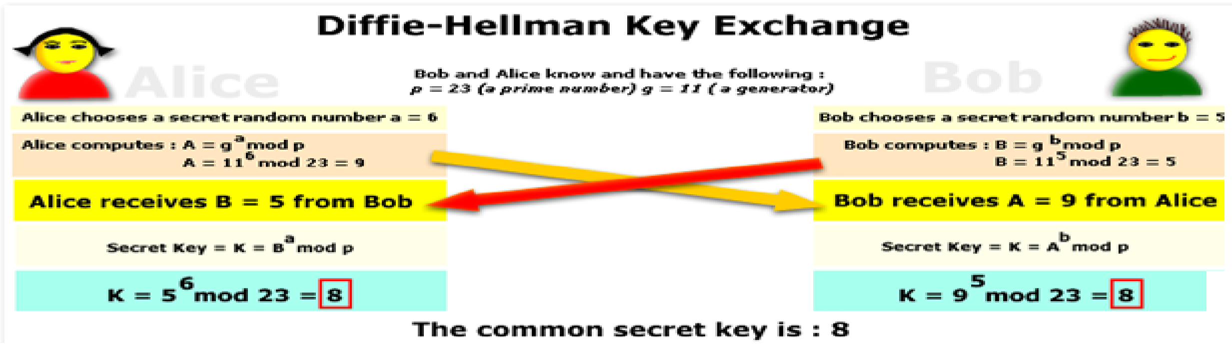
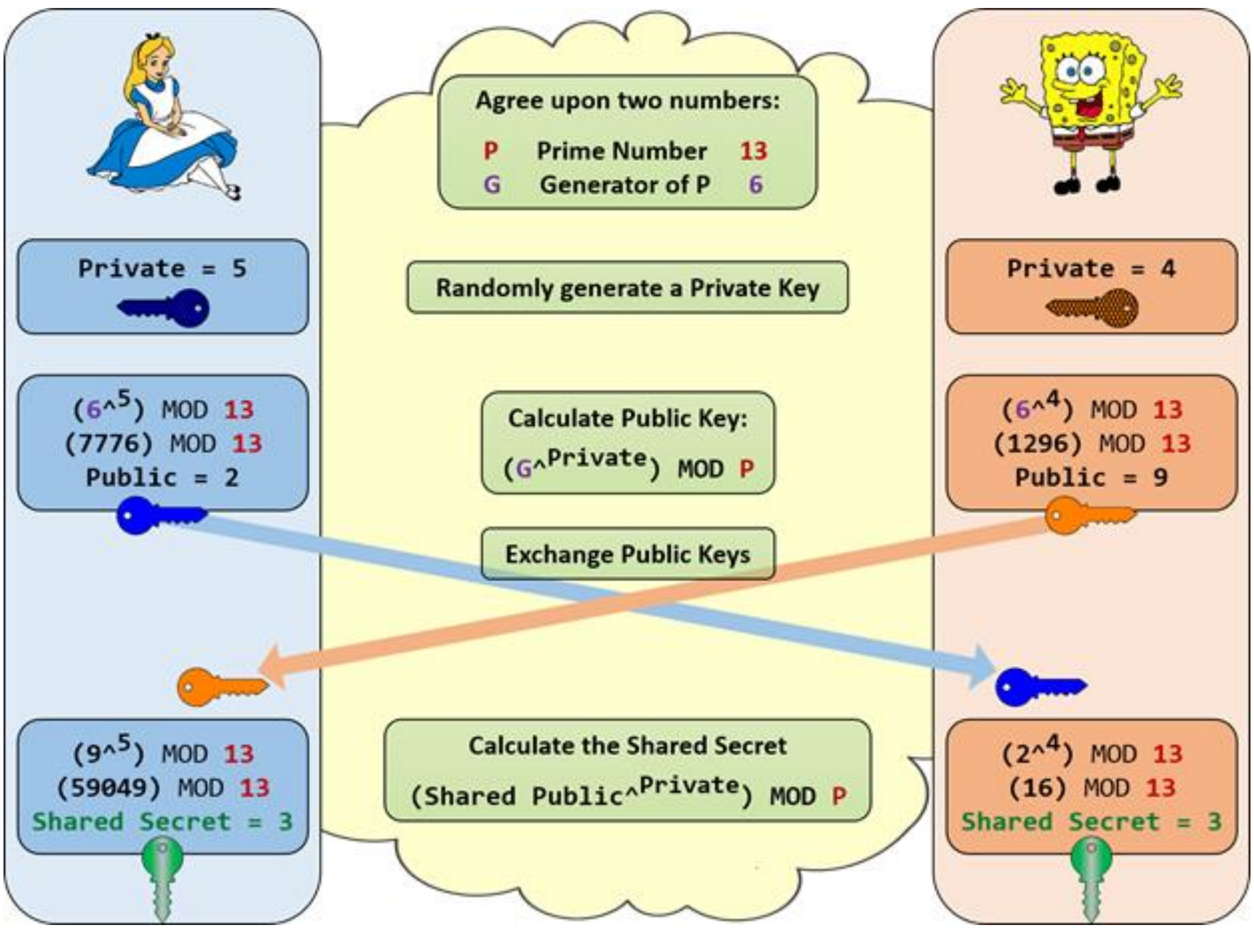
Diffie-Hellman (DH):

- Diffie-Hellman key agreement algorithm was developed in the Year 1976.
- Dr. Whitfield Diffie and Dr. Martin Hellman developed Diffie-Hellman Algorithm.
- Diffie-Hellman (DH) key exchange is a wonderful mathematical algorithm.
- Which allows two parties who have no prior knowledge to generate same secret keys.
- Diffie-Hellman allows two devices to establish a shared secret over an unsecure network.
- The encryption key for the two devices is used as a symmetric key for encrypting data.
- Diffie-Hellman key agreement algorithm is widely used in security protocols like IPSec.
- Diffie-Hellman algorithm is also use in Secure Shell (SSH) & Transport Layer Security (TLS).
- Only two parties involved in the DH key exchange and the key is never sent over the wire.
- Diffie-Hellman key group is a group of integers used for the Diffie-Hellman key exchange.
- Cisco Firewalls and Routers can use Diffie-Hellman (DH) groups 1, 2, 5, 14, 15, 19, and 20.
- Diffie-Hellman groups determine the strength of the key used in the key exchange process.
- Higher group numbers are more secure but require additional time to compute the key.
- Diffie-Hellman groups (DH) is used within IKE to establish session keys.
- 768-bit and 1024-bit D-H groups are supported in the Cisco routers and Firewall.
- The 1024-bit group is more secure because of the larger key size.
- In terms of VPN, it is used in the in IKE or Phase1 part of setting up the VPN tunnel.
- There are multiple, Diffie-Hellman Groups that can be configured in an IKEv2 policy.
- Both peers in VPN exchange must use same DH group, which is negotiated during Phase 1.
- There are multiple Diffie-Hellman Groups 1 to 30 assigned and 31-32767 Unassigned.

DH Group Number	Group Description	Recommendation
Diffie-Hellman group 1	768-bit Modulus	Avoid
Diffie-Hellman group 2	1024-bit Modulus	Avoid
Diffie-Hellman group 5	1536-bit Modulus	Avoid
Diffie-Hellman group 14	2048-bit Modulus	Acceptable
Diffie-Hellman group 15	3072-bit Modulus	Acceptable
Diffie-Hellman group 19	256-bit Elliptic Curve	Elliptic Curve Acceptable
Diffie-Hellman group 20	384-bit Elliptic Curve	Elliptic Curve NGE
Diffie-Hellman group 21	521-bit Elliptic Curve	Elliptic Curve NGE
Diffie-Hellman group 24	2048-bit Modulus	Next Generation Encryption


Diffie-Hellman Key Exchange
Alice
Bob
Bob and Alice know and have the following :
p = 23 (a prime number) g = 11 (a generator)
Alice chooses a secret random number a = 6
Alice computes : A = g^a mod p
A = 11^6 mod 23 = 9
Alice receives B = 5 from Bob
Secret Key = K = B^a mod p
K = 5^6 mod 23 = 8
Bob chooses a secret random number b = 5
Bob computes : B = g^b mod p
B = 11^5 mod 23 = 5
Bob receives A = 9 from Alice
Secret Key = K = A^b mod p
K = 9^5 mod 23 = 8
The common secret key is : 8


Agree upon two numbers:
P Prime Number 13
G Generator of P 6
Private = 5
Private = 4
Randomly generate a Private Key
(6^5) MOD 13
(7776) MOD 13
Public = 2
Calculate Public Key:
(G^Private) MOD P
(6^4) MOD 13
(1296) MOD 13
Public = 9
Exchange Public Keys
(9^5) MOD 13
(59049) MOD 13
Shared Secret = 3
Calculate the Shared Secret
(Shared Public^Private) MOD P
(2^4) MOD 13
(16) MOD 13
Shared Secret = 3